O_i and E_i are respectively the observed and expected frequency of ith class such that $\sum_{i=1}^{k} O_i = \sum_{i=1}^{k} E_i$.

If any of E_i is found less than 5, the corresponding class frequency may be pooled with preceding or succeeding classes such that E_i's of all classes are greater than or equal to 5. It may be noted that the value of k may be determined after pooling the classes.

The approximate sampling distribution of the test statistic under H_0 is the chi-square distribution with k-1-s *d.f* , s being the number of parametres to be estimated.

Step 5 : Calculation

Calculate the value of chi-square as

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

The above steps in calculating the chi-square can be summarized in the form of the table as follows:

Step 6 : Critical value

The critical value is obtained from the table of χ^2 for a given level of significance α.

Step 7 : Decision

Decide on rejecting or not rejecting the null hypothesis by comparing the calculated value of the test statistic with the table value, at the desired level of significance.

Example 2.11

Five coins are tossed 640 times and the following results were obtained.

Number of heads	0	1	2	3	4	5
Frequency	19	99	197	198	105	22

Fit binomial distribution to the above data.

Solution:

Step 1 : Null hypothesis H_0: Fitting of binomial distribution is appropriate for the given data.

Alternative hypothesis H_1: Fitting of binomial distribution is not appropriate to the given data.

Step 2 : Data

Compute the expected frequencies:

n = number of coins tossed at a time = 5

Let X denote the number of heads (success) in n tosses

N = number of times experiment is repeated = 640

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